

SEMI CONDUCTOR

- ↳ Have dual nature properties of Conductor & Insulator.
- Conductivity of semiconductor can be controlled.
- Direction & Magnitude of flow of charge can be controlled.
- Uni-directional current flow.

Examples

⇒ Elemental semiconductors

Eg → Si & Ge

⇒ Compound semiconductors

Eg → CdS, GaAs, CdSe, InP etc.

BAND THEORY

$$E_g = (E_{CB})_{\min} - (E_{VB})_{\max}$$

- CB → conduction band (free e^-)
- VB → valence band (valence e^-)

Types of Material

① Conductor

- CB & VB overlap
- $E_g = 0$
- $\sigma = 10^2$ to 10^8 S m^{-1}

② Semi-conductor

- $E_g < 3 \text{ eV}$
 - $\sigma = 10^5$ to 10^{-6} S m^{-1}
- Si → 1.1 eV
Ge → 0.7 eV

③ Insulator

- $E_g \geq 3 \text{ eV}$
- $\sigma = 10^{-11}$ to $10^{-19} \text{ S m}^{-1}$

INTRINSIC SEMI CONDUCTOR

n_e - conc. of free e^-

n_h - conc. of holes

n_i - Intrinsic carrier conc.

$$n_e = n_h = n_i$$

AT $T = 0 \text{ K}$ (Insulator)

⇒ In pure semiconductor

Temp ↑, no. of e^- hole pairs ↑
Conductivity ↑, Resistivity ↓

Hole (vacant space of e^-)

- $q = +ve$
- $(M_{eff})_{hole} > (M_{eff})_{e^-}$
- $\mu_h < \mu_e$ ($\mu = \frac{e\tau}{m}$)

Recombination (hole e^- pair)

↳ When free e^- collide with hole then ~~free~~ free e^- recombine with the hole that decreases the conductivity.

↳ Temp ↓ (e^- & h^+ K.E ↓)

Then e^- hole pairs will recombine

Conduction in pure semiconductor

- Motion of e^- hole is in opposite direction but current due to hole & e^- are in same direction.

$$I = I_e + I_h$$

$$\sigma = e[n_e \mu_e + n_h \mu_h]$$

$$I_e = n_e e A V_d \mu_e$$

$$I_h = n_h e A V_d \mu_h$$

$$I = \frac{n_i e A V_d [\mu_e + \mu_h]}{l}$$

EXTRINSIC SEMI CONDUCTOR

Doping → process of adding Impurity

IMPURITIES

① N-Type

- Pentavalent
- $n_e \gg n_h$
- Grp - 15 (P, As, Sb)
- Majority - e^-
Minority - holes
- Donor core is 1 unit (+ve) charge

② P-Type

- Trivalent
- $n_h \gg n_e$
- Grp - 13 (B, Al, Ga)
- Majority - holes
Minority - e^-
- Acceptor core is 1 unit (-ve) charge

LAW OF MASS ACTION

$$n_e \cdot n_h = n_i^2$$

N-Type

P-Type

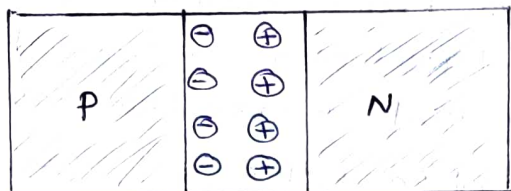
$$n_e \approx N_D (\text{Donor})$$

$$n_h \approx N_A (\text{Acceptor})$$

$$n_h = \frac{n_i^2}{n_e} = \frac{n_i^2}{N_D}$$

$$n_e = \frac{n_i^2}{n_h} = \frac{n_i^2}{N_A}$$

P-N JUNCTION DIODE



Depletion region

★ NOTE:-

- A depletion layer is formed at junction.
- Electric field formed from N to P side within Depletion region.
- width of depletion $\propto \frac{1}{\text{doping level}} \propto \text{Temp}^r$
- Drift current now from N to P due to electric field.

Initially, $i_{\text{diffusion}} > i_{\text{drift}}$

At equm, $i_{\text{diff}} = i_{\text{drift}}$

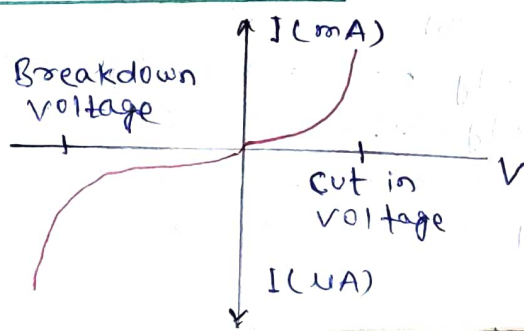
★ Barrier potential (V_B):-

for Si $\rightarrow 0.7V$, for Ge $\rightarrow 0.3V$

SEMICONDUCTOR DIODE

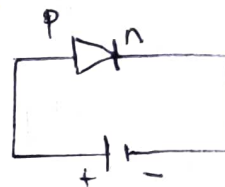
- It is a device having single P-n junction
- Symbol:- Anode Cathode

V-I characteristics



1 Forward Bias

- Depletion layer width $\downarrow$
- Effective barrier potential decreases



$$V_{\text{eff}} = V_{\text{applied}} - V_B$$

- low resistance at junction
- Current flow is of the order of mA.

2. Reverse Bias

- width of depletion layer $\uparrow$
- effective barrier potential increases
- $V_{\text{eff}} = V_{\text{applied}} + V_B$
- High resistance at junction
- Current flow is of the order of μA .
- Breakdown occurs at high reverse bias voltage.

Mx feel Table

Biasing	Electric field	Potential diff.	depletion width	$i_{\text{diffusion}}$	i_{drift}
forward	$\downarrow$	$\downarrow$	$\downarrow$	$\uparrow$	$\downarrow$
Reverse	$\uparrow$	$\uparrow$	$\uparrow$	$\downarrow$	$\uparrow$

Diode	Biasing
	R.B
	R.B
	F.B
	F.B
	R.B

Types of Breakdown

1) Avalanche

- Occurs in normal P-n junction diode
- low doping concn.
- High Reverse Voltage
- Material is damaged

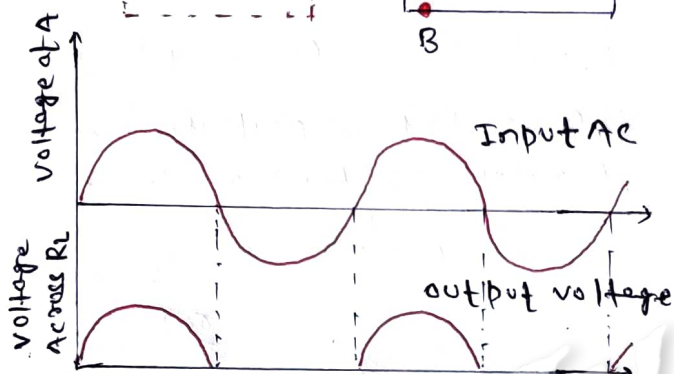
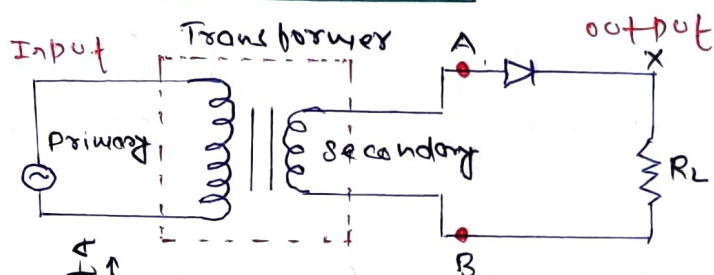
2) Zener

- Occurs in zener P-n junction diode.
- High doping conc.
- Low reverse voltage
- Material is not damaged.

Diode as Rectifier

- AC $\rightarrow$ DC

① Half Wave Rectifier



$$I_{rms} = \frac{I_0}{2}$$

$$I_{dc} = \frac{I_0}{\pi}$$

Efficiency,

$$\eta = \frac{(0.406) R_L}{R_L + R_f}$$

$R_L \rightarrow$ load resistance

$R_f \rightarrow$ forward resistance

$$\eta_{max} = 40.6\% \quad (R_f = 0)$$

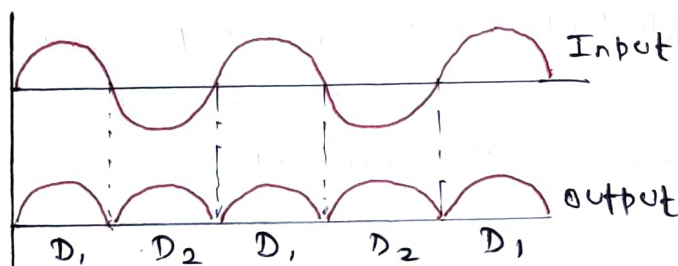
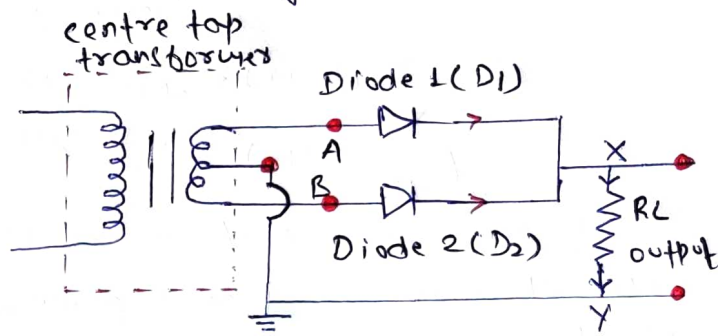
NOTE:-

- It rectifies either +ve or -ve cycle only of input signal.
- Frequency of output & input are same.

TG $\rightarrow$ @CYANNOTES

② Full wave Rectifier

- It rectifies both the cycle of the input.
- frequency of output is two times the frequency of input.



$$I_{rms} = \frac{I_0}{\sqrt{2}}$$

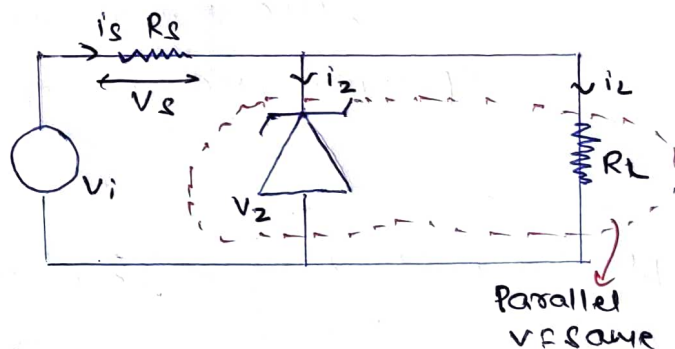
$$I_{dc} = \frac{2I_0}{\pi}$$

Efficiency,

$$\eta = \frac{(0.812) R_L}{R_L + R_f}$$

$$\eta_{max} = 81.2\% \rightarrow \text{for ideal diode}$$

Zener Diode as a Regulator

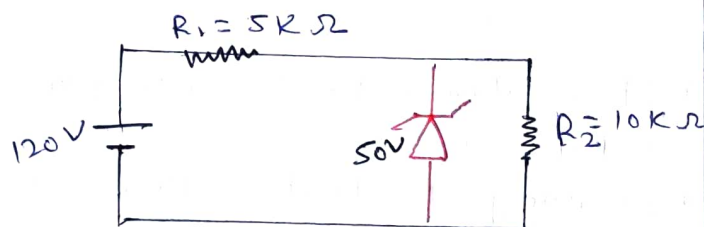


$$I_s = I_z + I_L \quad V_z = I_L R_L$$

$$I_s = \frac{V_s}{R_s} \quad V_i = V_s + V_z$$

$$V_{out} = V_z = I_L R_L = \text{constant}$$

Q $\rightarrow$ For the circuit shown below, the current through the zener diode is:-



Solⁿ Assuming zener diode does not undergo breakdown, current in circuit = $\frac{120}{15000} = 8 \mu A$.

Voltage drop across diode = 80 > 50V

The diode undergo breakdown.

Current in $R_1 = \frac{70}{5000} = 14 \mu A$

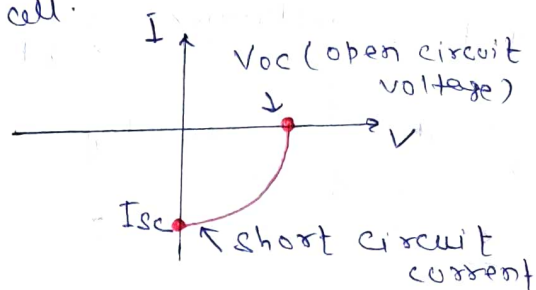
Current in $R_2 = \frac{50}{10000} = 5 \mu A$

Current through diode = 9 μA

Ga-As = Infrared
Ga-As-P = Red, yellow
Ga-P = Red, green

③ Solar cell

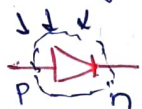
- P-N junction (under no bias) which generates EMF when irradiated with sunlight.
- Semiconductor with E_g closed to 1.5eV are ideal material for solar cell.



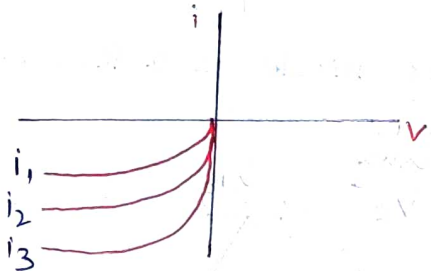
OPTOELECTRONIC DEVICE

① Photodiode

- Reverse biased P-n junction used for detecting optical signal.



$h\nu > E_g$



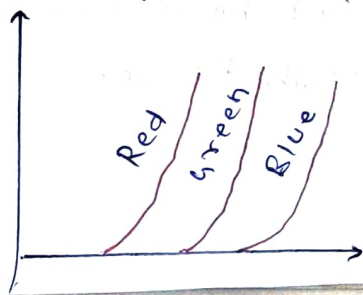
($i_3 > i_2 > i_1$) Intensity ↑
Photo current ↑

② LED (Light Emitting diode)



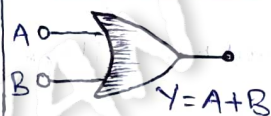
$h\nu \leq E_g$

- Heavily doped P-n junction used in forward bias for converting electricity to light.

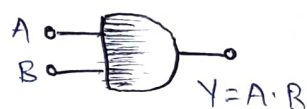


LOGIC GATES

① OR GATE



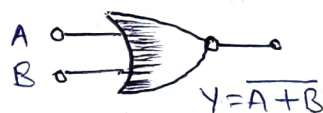
② AND GATE



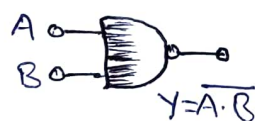
③ NOT GATE



④ NOR GATE



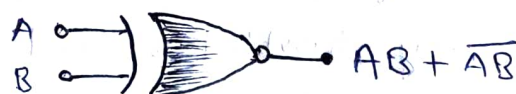
⑤ NAND GATE



Universal gates:- NOR & NAND

* Exclusive GATES

① XNOR



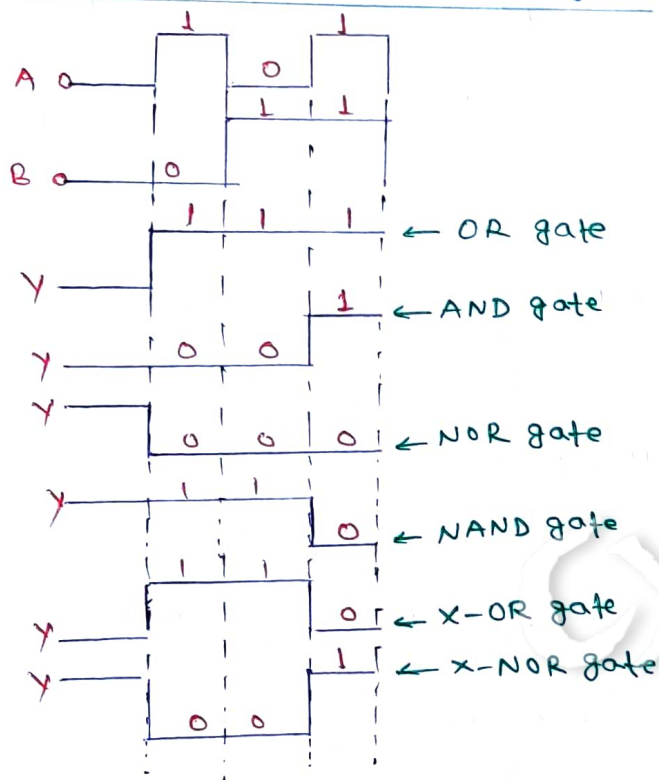
② XOR



Truth table of all GATES

A	B	OR	NOR	AND	NAND	X-OR	X-NOR
0	0	0	1	0	1	0	1
1	0	1	0	0	1	1	0
0	1	1	0	0	1	1	0
1	1	1	0	1	0	0	1

Time scale for different gates



Basic Boolean Exp:-

- $A + A = A$
- $A \cdot A = A$
- $A + 1 = 1$
- $A \cdot 1 = A$
- $\overline{\overline{A}} = A$
- $A + 0 = A$
- $A \cdot 0 = 0$
- $A \cdot \overline{A} = 0$
- $A + \overline{A} = 1$

De Morgan's principal:-

- $\overline{A + B} = \overline{A} \cdot \overline{B}$
- $\overline{A \cdot B} = \overline{A} + \overline{B}$

Special Case:-

- $\overline{\overline{A} + \overline{B}} = AB$
- $\overline{\overline{A} \cdot \overline{B}} = A + B$

* formation of different gates using NAND GATES

Gate	NOT	AND	OR	NOR
No. of NAND Gate req.	1	2	3	4

* formation of different gates using NOR gate

GATE	NOT	AND	OR	NOR
No. of NOR Gate req.	1	3	2	4

TH → @CYANNOTES

Thank you 😊

Best of luck for your
NEET EXAM

THE END!!